Setup & Data Upload

```
# ================================================
# Logistic Regression - Airline Customer Satisfaction
# ================================================

!pip install pandas seaborn scikit-learn matplotlib statsmodels -q

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, classification_report, roc_auc_score,
from sklearn.preprocessing import LabelEncoder, StandardScaler
import warnings
warnings.filterwarnings('ignore')

print("✅ Libraries imported!")

# Upload the dataset
from google.colab import files
print("📤 Please upload 'Invistico_Airline.csv'")
uploaded = files.upload()

filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
print(f"✅ Dataset loaded! Shape: {df.shape}")
df.head()
```

```
✅ Libraries imported!
📤 Please upload 'Invistico_Airline.csv'
```
Choose Files Invistico_Airline.csv

Invistico_Airline.csv(text/csv) - 11960454 bytes, last modified: 6/21/2026 - 100% done

```
Saving Invistico_Airline.csv to Invistico_Airline.csv
✅ Dataset loaded! Shape: (129880, 22)
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arriv time convenie
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0	
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0	
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0	
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0	
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0	

5 rows × 22 columns

Data Exploration & Cleaning

```
print("=== Dataset Info ===")
print(df.info())
print("\n=== Missing Values ===")
print(df.isna().sum())

# Handle missing values (common in Arrival Delay)
df_clean = df.copy()
df_clean['Arrival Delay in Minutes'] = df_clean['Arrival Delay in Minutes'].fillna(

# Target variable distribution
print("\nTarget Distribution (satisfaction):")
print(df_clean['satisfaction'].value_counts(normalize=True))
```

```
 #   Column                           Non-Null Count   Dtype
---  ------                           --------------   -----
 0   satisfaction                     129880 non-null  object
 1   Customer Type                    129880 non-null  object
 2   Age                              129880 non-null  int64
 3   Type of Travel                   129880 non-null  object
 4   Class                            129880 non-null  object
```

```
 9   Gate location                      129880 non-null  int64
 10  Inflight wifi service              129880 non-null  int64
 11  Inflight entertainment             129880 non-null  int64
 12  Online support                     129880 non-null  int64
 13  Ease of Online booking             129880 non-null  int64
 14  On-board service                   129880 non-null  int64
 15  Leg room service                   129880 non-null  int64
 16  Baggage handling                   129880 non-null  int64
 17  Checkin service                    129880 non-null  int64
 18  Cleanliness                        129880 non-null  int64
 19  Online boarding                    129880 non-null  int64
 20  Departure Delay in Minutes         129880 non-null  int64
 21  Arrival Delay in Minutes           129487 non-null  float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB
None

=== Missing Values ===
satisfaction                           0
Customer Type                          0
Age                                    0
Type of Travel                         0
Class                                  0
Flight Distance                        0
Seat comfort                           0
Departure/Arrival time convenient      0
Food and drink                         0
Gate location                          0
Inflight wifi service                  0
Inflight entertainment                 0
Online support                         0
Ease of Online booking                 0
On-board service                       0
Leg room service                       0
Baggage handling                       0
Checkin service                        0
Cleanliness                            0
Online boarding                        0
Departure Delay in Minutes             0
Arrival Delay in Minutes             393
dtype: int64

Target Distribution (satisfaction):
satisfaction
satisfied       0.547328
dissatisfied    0.452672
Name: proportion, dtype: float64
```

Feature Engineering & Encoding

```
# Encode categorical variables
categorical_cols = ['Customer Type', 'Type of Travel', 'Class']
df_encoded = pd.get_dummies(df_clean, columns=categorical_cols, drop_first=True)

# Encode target
le = LabelEncoder()
df_encoded['satisfaction'] = le.fit_transform(df_encoded['satisfaction'])  # satisf
```

```
# Select features (exclude ID-like columns if any)
feature_cols = [col for col in df_encoded.columns if col != 'satisfaction']
X = df_encoded[feature_cols]
y = df_encoded['satisfaction']

print(f"Features shape: {X.shape}")
print("Categorical encoding completed.")
```

```
Features shape: (129880, 22)
Categorical encoding completed.
```

Train-Test Split & Model Building

```
# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_st

# Scale numerical features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Build Logistic Regression model
model = LogisticRegression(max_iter=1000, random_state=42)
model.fit(X_train_scaled, y_train)

print("✅ Logistic Regression model trained!")
```

```
✅ Logistic Regression model trained!
```

Model Evaluation (Confusion Matrix, Precision, Recall)

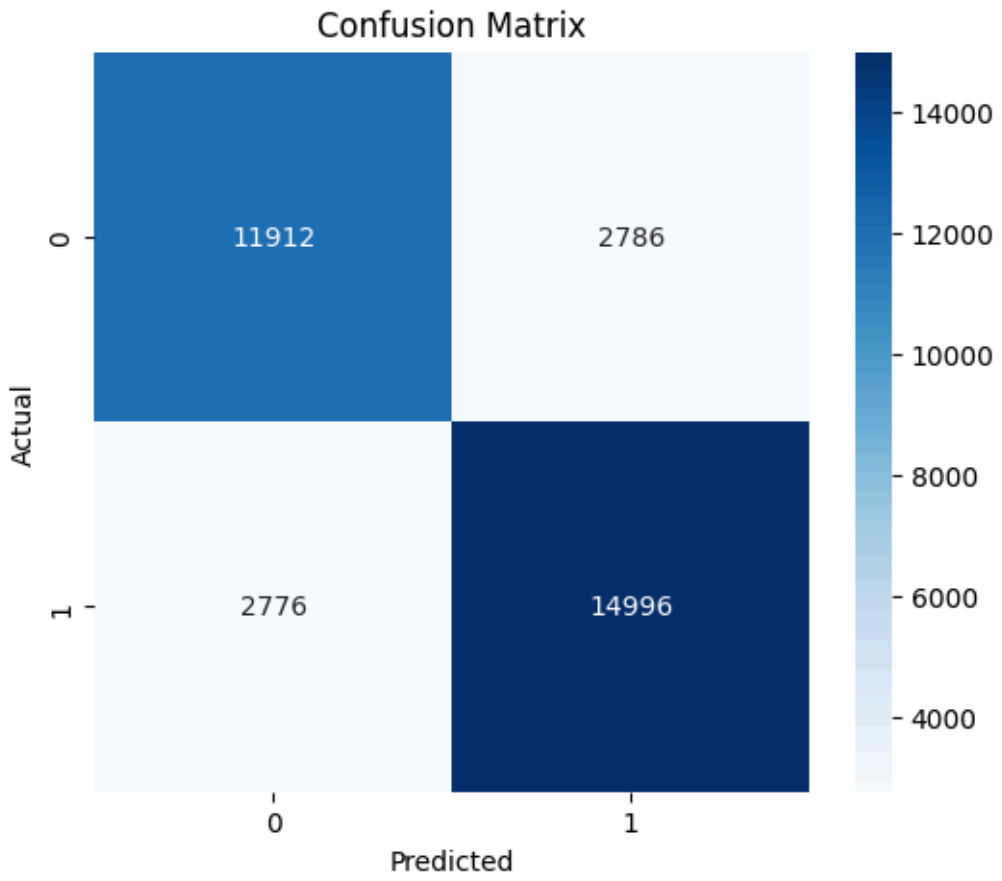
```
# Predictions
y_pred = model.predict(X_test_scaled)
y_pred_proba = model.predict_proba(X_test_scaled)[:, 1]

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title('Confusion Matrix')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()

# Classification Report
print("=== Classification Report ===")
print(classification_report(y_test, y_pred, target_names=['neutral or dissatisfied'

# ROC AUC
```

```
auc = roc_auc_score(y_test, y_pred_proba)
print(f"\nROC AUC Score: {auc:.4f}")
```




```
=== Classification Report ===
                         precision    recall  f1-score   support

neutral or dissatisfied       0.81      0.81      0.81     14698
              satisfied       0.84      0.84      0.84     17772

               accuracy                           0.83     32470
              macro avg       0.83      0.83      0.83     32470
           weighted avg       0.83      0.83      0.83     32470


ROC AUC Score: 0.9028
```

Coefficient Interpretation & Business Insights

```
# Feature importance (coefficients)
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_[0]
}).sort_values(by='Coefficient', ascending=False)

print("=== Top Positive Drivers of Satisfaction ===")
print(coefficients.head(8))

print("\n💼 BUSINESS RECOMMENDATIONS:")
print("• Focus on improving **Inflight entertainment**, **Seat comfort**, and **Onli
```

```
print("• Reduce **Arrival Delay** as delays significantly decrease satisfaction proba
print("• Business travelers and loyal customers show different patterns — tailor serv
print("\nModel Performance: High Precision & Recall for Satisfied customers.")
```

```
=== Top Positive Drivers of Satisfaction ===
                   Feature  Coefficient
7   Inflight entertainment     0.971217
10        On-board service     0.393796
2              Seat comfort     0.390712
13         Checkin service     0.356484
9   Ease of Online booking     0.335298
11        Leg room service     0.310157
15         Online boarding     0.183747
5            Gate location     0.165566

💼 BUSINESS RECOMMENDATIONS:
• Focus on improving **Inflight entertainment**, **Seat comfort**, and **Online supp
• Reduce **Arrival Delay** as delays significantly decrease satisfaction probability
• Business travelers and loyal customers show different patterns — tailor services a

Model Performance: High Precision & Recall for Satisfied customers.
```